

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

1. Solve the recurrence  $T(n) = 5T(n/3) + n^2$ .

Justify your answer showing how you derived your solution.

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**Answer:**

1. We apply the master theorem:

(a)  $a = 5$ ,  $b = 3$ ,  $f(n) = n^2$ .

(b)  $\log_b(a) \approx 1.46$ , case 3 applies.

(c) Take  $\epsilon = 0.1$ , then  $f(n)$  is  $\Omega(n^{1.56})$ ,  
and  $af(n/b) = 5(n/3)^2 = (5/9)n^2$ , and  $c = 5/9 < 1$ .

Therefore,  $T(n)$  is  $\Theta(n^2)$ .